Starting Finance Club PoliTo

Research Paper

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RESEARCH ARTICLE

Operating-Value Sensitivity for Barrick Mining

Option-implied gold and copper laws in a controlled operating experiment

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JEL classification
G12; G13; G31; C32; C53; C61; C63

Discoveries

Median proxy under each gold law, common operating contract.

Black–Scholes	56.32
Heston	57.62
Bates–Poisson	59.48
Bates–Hawkes	57.26

USD billion; signed terminal value. Monte Carlo range of one model: 56.24–59.99. Not company value.

Abstract

We ask how much the choice of gold-price model matters for a discounted operating-margin proxy of Barrick when production, costs and discount-rate shocks are held fixed. Black–Scholes, Heston, Bates and Bates–Hawkes are calibrated to 64 GLD calls of 2 September 2026, with the trust's 0.40% sponsor fee treated as carry, and drive daily gold paths from the spot price at the valuation time. Only cash flows after the valuation date are counted. Across 8,192 paths the median proxy lies between USD 56.32 and 59.48 bn for all four models, a spread no larger than the numerical sensitivity of a single model (USD 56.24–59.99 bn across path counts and seeds). The terminal value accounts for about 83% of the mean. In 30 rolling repricing tests Bates–Hawkes has the lowest error, but no model beats the persistence of the implied-volatility surface. Three producing copper mines add about USD 2.30–2.33 bn of explicit flows to 2030. The proxy has no equity bridge; it is a controlled research sensitivity, not a fair value, target price or investment recommendation.

RESEARCH PROGRAMME AND COLLABORATION

This paper grew out of an eight-team project involving 17 students with complementary backgrounds in mathematics, data science, engineering, economics and finance. The size of the author group reflects a division of research responsibilities and a shared learning process: Teams 1–3 developed the mathematical, econometric and simulation foundations; Teams 4–5 built the operating and corporate-valuation layers; and Teams 6–8 studied gold-price dynamics, market evidence and option-implied calibration.

The extended technical report records definitions, derivations and implementation choices to establish a common technical language across these backgrounds. Teams worked on dedicated modules, then compared assumptions, revised their contributions and reconciled dates, units and interfaces before integration. The present paper condenses this work into a single auditable experiment. Together with the reproducible code, it demonstrates the group's existing research practice and provides a basis for developing a sustained student-led quantitative research team with academic guidance.

1 INTRODUCTION

A gold producer's value depends on the distribution of future commodity prices as well as its operating and financing assumptions. A single price path can hide this source of model risk. We ask how an unconstrained Barrick operating-proxy distribution changes when progressively richer gold-price processes are substituted within a common operating scenario and discounted-margin contract.

We compare Black–Scholes, Heston, Bates–Poisson and Full Bates–Hawkes calibrated to GLD options. Current-surface diagnostics and rolling out-of-sample tests distinguish fit from predictive performance; shared production, costs and WACC shocks isolate the effect of the gold engine. The contribution is this controlled comparison and its auditable integration, rather than a claim that any model provides an unconditional fair value.

2 DATA AND VALUATION SETUP

2.1 Controlled layer separation

The integrated experiment is organized as a controlled handoff across three modules. Team 8 governs only the gold-price process; Team 4 supplies the operating forecasts; and Team 5 maps those inputs through a common tax, WACC and terminal-value contract without a reconciled equity bridge. In the four-model comparison only the gold engine changes: the same Team 4 production and unit-cost forecasts, the same Team 5 corporate rules and the same WACC shocks are reused across runs.

2.2 Valuation clock

The valuation time is the 2 September 2026 close (20:00 UTC). Only flows after it are valued. Barrick's Q1–Q2 2026 actuals have already been realised and are excluded; they are used only to check levels. The quarter that contains the valuation date contributes its remaining 28 of 92 days, under the stated assumption that production and unit cost are uniform within the quarter. Eighteen quarters follow, to 31 December 2030 (4.33 years). Revenue in each quarter uses the average simulated daily gold price over its remaining interval, and each flow is discounted from the midpoint of that interval.

2.3 Calibration snapshot and carry

The calibration uses the 2 September 2026 GLD call surface acquired through the IB API. After the common DTE $\geq$ 75 and numerical filters, the dense surface contains 605 eligible observations across 12 expiries and 146 strikes, with 79–653 days to expiry. The 8×8 Chebyshev–Chebyshev sample contains 64 distinct actual contracts spanning eight expiries and 20 strikes. The rate input is a same-date NSS fit to 14 U.S. Treasury tenors (Svensson, 1994; U.S. Treasury, 2026), with a 2.0608 bp RMSE against the continuously compounded par-yield proxy; it is not a bootstrapped zero curve.

GLD's sponsor fee of 0.40% a year reduces the gold backing of each share, so the GLD forward is $Se^{(r-q)T}$ with $q = 0.40\%$ even though no cash is distributed (SPDR, 2025). Two independent observations support this value: the GLD/gold log-ratio trend of −0.396% a year over 2022–2026, and COMEX gold futures, whose carry against the

synchronous spot implies a GLD yield of 0.26–0.49% at 0.5–0.8 years. The carry is applied exactly through $C(S,K;q) = e^{-qT}C(S,Ke^{qT};0)$.

The gold scenarios start from the XAUUSD spot of USD 4,391.32/oz at the valuation time, synchronous with the GLD close. The earlier draft started from Barrick's Q2 average realised price of USD 4,417/oz, which predates the valuation by two months.

2.4 Probability measure

Option parameters are estimated under $\mathbb{Q}$. The scenario drift is the integral-preserving forward curve of the Treasury fit. Read as a physical expectation, this assumes no gold risk premium, which is a common assumption for an asset with near-zero market beta but is not estimated here. The option-implied volatility shape is carried over as a conditional assumption; it is not a validated physical law. Expected flows are then discounted at WACC, as in an ordinary DCF. For comparison, Section 7 also reports the explicit flows discounted at the risk-free curve. A full $\mathbb{Q} \to \mathbb{P}$ specification would need price, variance and jump risk premia (Section 3.4).

Table 1. Contract held fixed across gold engines.

Block	Current experiment
Gold layer	Black–Scholes/GBM, Heston, Bates–Poisson and Full Bates–Hawkes calibrated by Team 8 under $\mathbb{Q}$ with $q = 0.40\%$.
Operations	Team 4 production and unit-cost forecasts from 2026Q3 (stub) to 2030Q4; actuals excluded.
Corporate mapping	Team 5 tax, growth–ROIC–reinvestment schedule, stochastic WACC and explicit terminal convention; no equity bridge.
Simulation	8,192 paths, daily steps, one common WACC shock array.

2.5 Option exercise convention

GLD contracts are American-style (Cboe, 2026). With $q = 0.40\%$ below the Treasury rate, early exercise of a call is not optimal in the frictionless model. An 800-step CRR check on all 64 contracts gives a maximum American–European premium of USD 2.3e-10; this tests the convention, not market frictions.

2.6 Operating-model status

Team 4's original operating study is richer than the object passed downstream. Cost of Sales is forecast from a log-space master chain; production is constructed bottom-up from ore processed, average grade and recovery. The valuation uses the resulting point forecasts, so mine-level operating uncertainty is not propagated jointly with gold-price uncertainty. Team 4's legacy price simulation and illustrative EBITDA/valuation output are excluded.

2.7 Gold perimeter and reporting basis

The Team 4 subset comprises Bulyanhulu, Carlin, Cortez, Kibali, Loulo–Gounkoto, North Mara, Pueblo Viejo and Turquoise Ridge. Its quarterly estimation window is 2019Q1–2025Q4, with North Mara beginning in 2019Q4. The operating quantities are already attributable: Nevada Gold Mines is 61.5%, Pueblo Viejo 60%, Kibali 45%, Loulo–Gounkoto 80%, and the Tanzanian mines 84%. Kibali is equity-accounted, whereas the other interests are consolidated with partners' non-controlling interests (Barrick, 2025). Those partners must not be deducted again from attributable flows. Tanzania's contractual economic-benefit sharing still requires a separate cash-flow reconciliation.

The forecast covers the eight-mine subset, which produced 622 and 711 koz in Q1–Q2 2026 against company-wide totals of 719 and 796 koz. The proxy therefore omits the remaining gold mines. Cost of Sales includes depreciation and price-sensitive royalties; multiplying its sold-ounce cost by production gives a component-margin proxy, not EBITDA or reconciled EBIT. Loulo–Gounkoto's 2025Q4 accounting uplift also makes its cost anchor a normalization issue. Ownership, the North America IPO, cash and debt, share count and reserves must be fixed at the valuation date before any per-share result can be reported.

3 METHODOLOGY

3.1 Gold-price engines

The four Team 8 specifications relax one restriction at a time. Under $\mathbb{Q}$, Black–Scholes/GBM uses constant volatility,

$$\frac{\mathrm{d}S_t}{S_t} = (r_t - q_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t. \tag{1}$$

GBM is the constant-volatility benchmark. Heston introduces mean-reverting stochastic variance (Heston, 1993),

$$\frac{\mathrm{d}S_t}{S_t} = (r_t - q_t)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S, \tag{2}$$

$$\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v, \qquad \mathrm{d}W_t^S\,\mathrm{d}W_t^v = \rho\,\mathrm{d}t. \tag{3}$$

Bates adds compensated compound-Poisson jumps (Bates, 1996),

$$\frac{\mathrm{d}S_t}{S_{t^-}} = (r_t - q_t - \lambda k)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S + (e^J - 1)\,\mathrm{d}N_t, \tag{4}$$

where $k = \mathbb{E}^{\mathbb{Q}}[e^J - 1]$.

3.2 Full Bates–Hawkes: implemented affine engine

The full engine preserves Heston variance but replaces the constant Poisson arrival rate with an exponential Hawkes state (Hawkes, 1971):

$$\frac{\mathrm{d}S_t}{S_{t^-}} = (r_t - q_t - \lambda_t k)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S + (e^J - 1)\,\mathrm{d}N_t, \tag{5}$$

$$\mathrm{d}v_t = \kappa(\theta_v - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v, \tag{6}$$

$$\mathrm{d}\lambda_t = \beta(\bar{\lambda} - \lambda_t)\,\mathrm{d}t + \alpha\,\mathrm{d}N_t, \qquad n = \alpha/\beta < 1. \tag{7}$$

Under the implemented exponential-kernel specification, pricing exploits the affine Markov structure rather than a standalone finite-difference PIDE. Writing $X_t = \log S_t$, the characteristic function factorizes into Heston diffusion and Hawkes jump blocks,

$$\phi_{X_T}(u) = \phi_{\mathrm{diff}}(u)\exp(A(u,T) + B(u,T)\lambda_0)\,, \tag{8}$$

with the jump block determined by the Riccati system

$$\frac{\mathrm{d}B}{\mathrm{d}\tau} = \psi_J(u)e^{\alpha B} - \beta B - iuk - 1, \tag{9}$$

$$\frac{\mathrm{d}A}{\mathrm{d}\tau} = \beta\bar{\lambda}B, \qquad A(0) = B(0) = 0, \tag{10}$$

where $\psi_J(u) = \exp(iu\mu_J - \frac{1}{2}\sigma_J^2u^2)$. European call prices use Fourier-COS inversion (Fang and Oosterlee, 2008), with independent diffusion and jump drivers. The identities $\phi(0) = 1$, the martingale condition at $u = -i$, and the limit $\alpha \to 0$, $\lambda_0 = \bar{\lambda}$, back to Bates are used as implementation checks.

Model-role constraint

These processes govern commodity prices only. The copper extension uses forward-price analogues; neither price layer generates Barrick production or costs.

3.3 Option-surface calibration and predictive test

Every model is fitted to the same structured GLD surface by minimizing a vega-scaled price loss,

$$F(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{C_i^{model}(\theta) - C_i^{mkt}}{\max(\mathrm{Vega}_i, \varepsilon)}\right]^2, \tag{11}$$

with Vega per unit decimal volatility and $\varepsilon = 10^{-4}$. This is price-based MSE; exact IV RMSE uses inversion, a square root and 10^4 scaling. Nested starts are retained during SLSQP refinement. A Feller strict-positivity constraint is imposed by choice; Hawkes stability requires $n < 1$.

Calibration fit and forecasting are evaluated separately. A date is admitted to the rolling test only when at least 64 eligible observations and three expiries survive the common filters. Each CC64 origin is scored on the next dense date. With n_t common targets on date t, equal-date performance is

$$R^2_{OOS,m|b} = 1 - \frac{\sum_t n_t^{-1}\sum_j e^2_{m,t,j}}{\sum_t n_t^{-1}\sum_j e^2_{b,t,j}}, \tag{12}$$

against both the origin cross-sectional mean IV and an origin observed-IV surface interpolated in maturity and log-moneyness inside its convex hull. The target spot and rate curve enter only as scoring-date conditioning variables. This design prevents the lowest in-sample IV error from being interpreted automatically as predictive dominance.

Origin parameters are frozen before next-date scoring; target quotes do not select them. Conditioning on target spot and rates makes this a repricing test, not a joint ex ante forecast. Persistence is evaluated only on its interpolation support; unmatched observations are not zero losses.

3.4 From pricing measure to physical revenues

Option coefficients under $\mathbb{Q}$ do not identify physical commodity returns. All four families can be estimated under $\mathbb{P}$, but their cross-measure link requires underlying history and identified risk premia (Pan, 2002). With $\mathrm{d}W^{Q} = \mathrm{d}W^{P} + \eta\sqrt{v}\,\mathrm{d}t$, the restricted variance map is $\kappa_P = \kappa_Q - \xi\eta_v$ and $\kappa_P\theta_P = \kappa_Q\theta_Q$. A positive intensity scaling c_I and normal mark tilt give $\lambda_P = c_I\lambda_Q$ and $\mu_{J,P} = \mu_{J,Q} + b_J\sigma_J^2$. Thus physical expected price return is

$$\mu_t^P = a_t^Q + \eta_S v_t + \lambda_t^Q(c_I k_P - k_Q), \tag{13}$$

where a_t^Q is pricing carry and $k_M = \mathbb{E}^M[e^J - 1]$. Hawkes baseline, initial intensity and excitation scale by c_I, while decay is unchanged; physical branching must stay below one. Algebraic closure does not identify premia. For the restricted CIR and unmarked linear Hawkes family, a stopped-density entropy bound establishes equivalence on finite horizons; its full assumptions and derivation are in the unified study (Schönbucher, 2001; Sokol and Hansen, 2015). Empirical estimation and predictive validation remain separate. In particular, copper Bates has zero fitted Q jump intensity: an equivalent change cannot create physical jumps without a new joint specification.

The 6 October diagnostic, still cut off at 2 September, adds 1,253 GLD regular-session prices and 1,003 synchronous GLD–gold observations, including early closes. Seven copper deliveries provide 408 accepted origin-selected, same-delivery returns; four requested earlier deliveries were unavailable. This avoids artificial rollover jumps but does not identify copper spot carry or realised selling prices. The approximately 100-contract option budget is separate from the daily time-series sample.

A training-only GLD GBM with an origin-fitted gold basis produces quarterly gold-price RMSE of USD 417.19/oz over 689 overlapping targets, versus USD 428.52/oz for a no-change mean. Its nominal 95% coverage is only 89.6%; direct gold GBM has better absolute error and log score. This is a physical historical benchmark, not a joint option/history fit. Preliminary fixed-current-Q Heston, Bates and Hawkes premium estimates remain sensitive to particle and time-step resolution and are not admitted to valuation.

Option repricing OOS, physical commodity OOS and Barrick operating OOS are distinct tests. Current pricing coefficients cannot be used at earlier physical forecast origins. The intended DCF requires $\mathbb{P}$ realised selling prices. Thirty recovered historical surfaces, each with 64 contracts, permit new dated calibrations. A conditional two-stage estimator may fit $\mathbb{Q}$ options and then physical premia using origin-available returns; it is not joint maximum likelihood and must address uncertainty from both stages. Forward paths share the option-estimated initial state across measures. All four physical models now have 29 common next-session targets. After origin-only gold conversion, RMSE is USD 66.35/oz for GBM, 66.77 for Heston, 66.84 for Bates and 65.81 for Bates–Hawkes, versus 66.09 for no-change. The small Hawkes advantage does not establish superiority. Nominal 95% intervals cover 27/29 targets for GBM and Heston and 25/29 for both jump models (binomial $p = 0.055$). At 80% the jump models under-cover significantly: Bates covers 17/29 ($p = 0.008$) and Bates–Hawkes 16/29 ($p = 0.004$), against 21/29 for GBM and 20/29 for Heston. Their central intervals are too narrow, and parameter and basis uncertainty are omitted. These origin fits predate the carry correction and use $q = 0$; they were not recalibrated. Two Heston fits and one Hawkes fit fail to converge and remain included; independent filter checks show substantial seed sensitivity. Resumed fits use strictly earlier parameters as initial guesses with global fallback, while preserving prior cold fits and recording the initialization date/hash. This short panel does not validate long-horizon physical distributions. Analytically integrating normal jump marks reduces seed dispersion in a targeted frozen-parameter check, but substantial sensitivity remains; the reported panel uses the original filter. A fixed-delivery copper future cannot be extended beyond expiry as a spot forecast, and a September valuation must exclude elapsed quarterly cash flows. The operating bridge also requires attributable sales, reconciled FCFF and compatible discounting. WACC does not convert $\mathbb{Q}$ paths to $\mathbb{P}$; frozen scenario values below remain a separate experiment. The unified study gives the complete coefficient derivation, data exclusions, numerical checks and daily/quarterly baseline results.

3.5 Team 4 cost and production models

For costs, Team 4 first builds a log-space global master chain from mine-level series and then selects an **ARIMA(3,1,0) with drift** by automated information-criterion search. The admitted specification is

$$(1 + 0.9501B + 0.6128B^2 + 0.3003B^3)(1 - B)y_t = 0.0222 + \varepsilon_t. \tag{14}$$

The selected model minimizes both $AIC = -94.64$ and $BIC = -86.87$ (Akaike, 1974; Schwarz, 1978), where p counts parameters and N observations,

$$AIC = 2p - 2\log\widehat{L}, \qquad BIC = p\log N - 2\log\widehat{L}. \tag{15}$$

The source's 12-quarter cost hold-out reports log-scale RMSE of 0.0504 against 0.1512 for a random walk. Its 0.603% MAPE is computed on log values, not USD/oz; it must not be read as a cost-level percentage error. This is source evidence, not independent replication.

Team 4 models production components without claiming AIC/BIC selection. With ore in kt, grade in g/t and recovery as a fraction, physical output in koz is

$$Q_t^{phys} = \mathrm{Ore}_t\,\mathrm{Grade}_t\,\mathrm{Recovery}_t/31.1035, \tag{16}$$

with ore modelled as $SARIMA(1,0,1)(1,0,0)_4$, grade as a non-seasonal AR(1), and recovery sampled uniformly from historical extrema widened by one percentage point and truncated to $[0.50, 1.00]$. Team 4 additionally applies a 0.91 empirical realization factor, not a unit conversion; its transfer across mines remains unvalidated. The valuation retains the frozen Team 4 vectors, not the Goodman distributions; production bands are 80% and cost bands 95%, while product-variance formulas require explicit dependence assumptions.

3.6 Path simulation

All four engines are simulated on a daily grid from the valuation time with full-truncation Euler variance (Lord et al., 2010): the variance state is carried unclipped and only $\max(v, 0)$ enters drift and diffusion. For Bates–Hawkes the intensity decays exactly between days, the number of jumps in a day is Poisson with the exact integral of that decay, and the drift is compensated by the same integral, so each step is a martingale given its starting state. Monte Carlo call prices at one and three years agree with the COS prices within about two batch standard errors for every model (8 independent batches of 16,384 paths).

3.7 Valuation mapping

Team 5 makes growth consistent with reinvestment through

$$b_k = \frac{g_k}{ROIC_k}, \qquad F_j^{proxy} = M_j^{AT} - \max(M_j^{AT}, 0)\, b_{k(j)}, \tag{17}$$

where M_j^{AT} is the after-tax component margin of quarter j and $k(j)$ the year since valuation in which its flow falls. Over the five rate years g_k declines linearly from 8.50% to 2.50%, $ROIC_k$ from 14.50% to 10.50%, and the reinvestment rate from 58.62% to 23.81%.

The common discount rate is mean reverting by year since valuation,

$$w_{k+1} = \bar{w} + (w_k - \bar{w})e^{-\kappa_w} + \sigma_w^*\varepsilon_{k+1}, \tag{18}$$

with $w_0 = 9.10\%$ applying to the first year, $\bar{w} = 8.30\%$, $\kappa_w = 0.45$ and annual volatility 0.90%. It is clipped to $[g_\infty + 1.00\%, 25.00\%] = [3.50\%, 25.00\%]$; every gold engine receives the same shock array. The discount factor $D(t)$ compounds $1 + w_k$ over the elapsed part of each year.

For path i, with flow times t_j at the midpoint of each quarter's remaining interval and horizon H at 31 December 2030,

$$V_{op}^{(i)} = \sum_j F_j^{proxy,(i)} D^{(i)}(t_j) + TV_H^{(i)} D^{(i)}(H). \tag{19}$$

The terminal block uses $g_\infty = 2.50\%$, $ROIC_\infty = 10.50\%$ and therefore $b_\infty = 23.81\%$, applied to the after-tax margin M_H^{AT} of the last four quarters:

$$TV_H^{(i)} = \frac{M_H^{AT,(i)}(1+g_\infty)(1-g_\infty/ROIC_\infty)}{w_H^{(i)} - g_\infty}. \tag{20}$$

The rate floor keeps the spread positive. The signed terminal keeps negative final margins; $TV = 0$ gives an explicit-flows sensitivity. Neither closure is a reserve-calibrated mine life or an optimal abandonment rule. No equity or share bridge is inferred.

4 CURRENT OPERATING-MODEL EVIDENCE

Team 4 supplies the inherited operating scenarios, checked against Q1–Q2 2026 statistics. Its earlier gold simulation is excluded: all price paths come from Team 8. These fixed inputs permit model attribution but are not independently validated operating forecasts.

4.1 Cost-of-sales block

The cost model in Section 3.5 extracts a common log-space trend, forecasts it with ARIMA(3,1,0) with drift, and reconstructs mine-level paths with volatility-scaled anchors. Its reported selection and hold-out diagnostics are inherited evidence, not revalidated forecasts or a jointly simulated cost state.

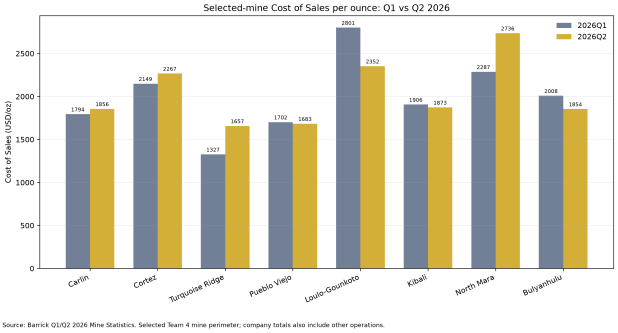


Figure 1. Selected-mine Cost of Sales per ounce in Q1 and Q2 2026. Mine dispersion motivates preserving the Team 4 operating decomposition rather than replacing it with a single company-average cost.

The valuation uses the 18 forecast cost entries from 2026Q3 to 2030Q4; the two 2026 actuals precede the valuation date and are not valued. It does not propagate the source study's operating distributions.

4.2 Production block

Production is reconstructed from ore processed, grade and recovery using the specifications in Section 3.5. This preserves mine heterogeneity. The SARIMA orders are admitted from the source study, which does not report their selection by AIC/BIC.

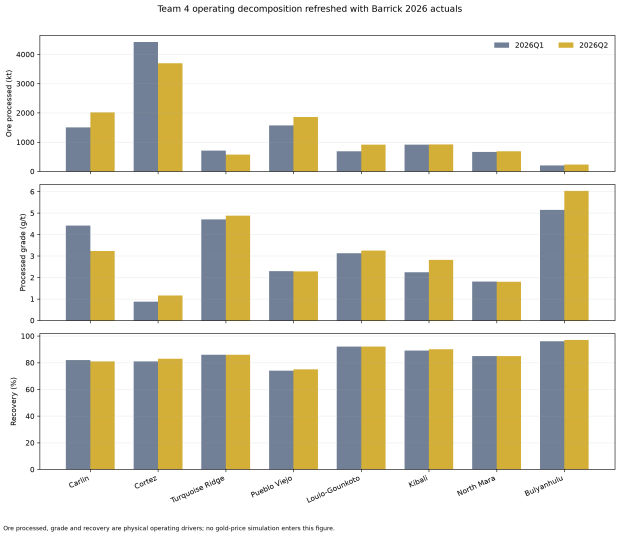


Figure 2. Ore processed, grade and recovery updated with Barrick 2026 actuals. These are physical operating drivers; no commodity-price simulation enters the panel.

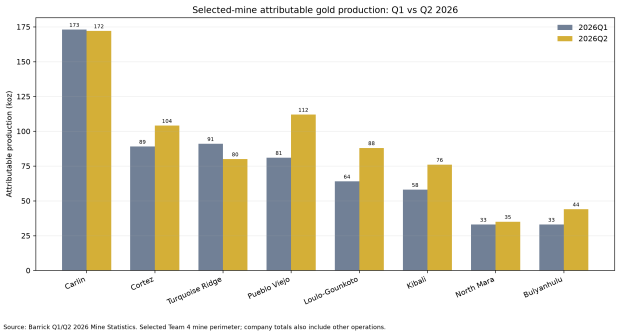


Figure 3. Selected-mine attributable gold production in Q1 and Q2 2026. The current actuals reveal heterogeneous changes that are hidden by a company-level total.

The original study also develops Goodman variance propagation and confidence bands for the production components. Those distributions remain useful evidence on operating-model uncertainty, but they are not propagated jointly in the four-engine corporate experiment: only the refreshed point vectors are passed downstream.

4.3 Forecast handoff used by the valuation

The final handoff keeps observed and modelled quantities distinct. Barrick's Q1–Q2 2026 actual production and CoS describe the operating state before the valuation date and are not discounted; the Team 4 forecasts begin at Q3 2026, of which only the part after 2 September is valued, and run to 2030Q4.

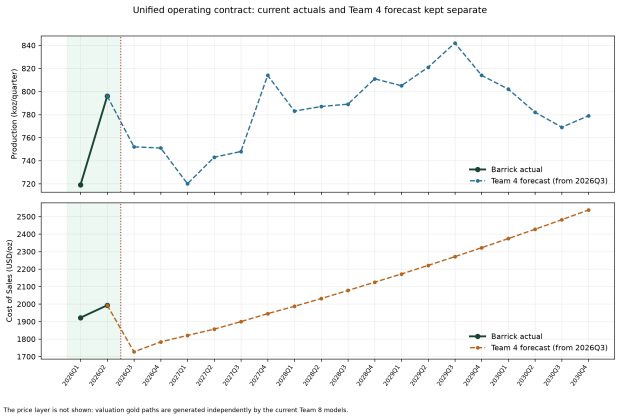


Figure 4. Inherited operating contract. Company-wide Q1–Q2 2026 actuals precede the eight-mine Team 4 forecast from Q3; only the forecast after 2 September enters the valuation, and the scope difference is discussed in the data section. The straight cost path is the conditional mean of a persistent-trend specification, not a claim that operating uncertainty disappears.

Prudential operating boundary

Option-implied laws supply conditional scenario shapes to the proxy layer. Costs retain a persistent trend without additional technology-driven cost reductions. This is prudential relative to otherwise identical efficiency-improvement scenarios, but it does not establish a mathematical lower bound. Stochastic costs could yield favourable and adverse outcomes; the current fixed vectors preserve attribution across gold engines. Growth is already included in the common DCF schedule.

5 CALIBRATION SAMPLE AUDIT

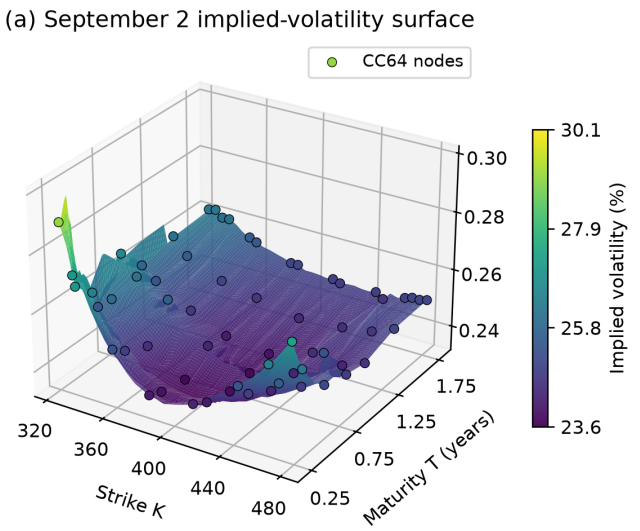


Figure 5. (a) GLD implied-volatility surface, 2 September 2026, with CC64 nodes. Interpolation is used for display only.

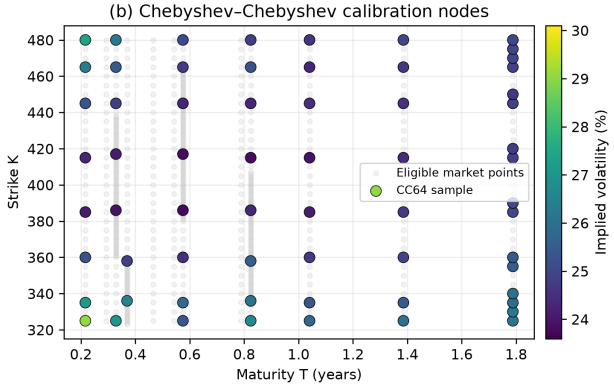


Figure 5 (continued). (b) The 64 actual contracts selected from 605 eligible calls across 12 expiries and 146 strikes.

5.1 Why structured sampling is retained

The option surface is irregular in strike and maturity. Team 8 therefore passes a fixed 8×8 Chebyshev–Chebyshev subset to the optimizer rather than allowing densely listed regions to dominate the loss. Every target is mapped to the nearest still-unused actual option: the 64-point calibration set contains no synthetic contracts. The figure audits the same 2 September parameter snapshot used by the refreshed valuation.

Evidence status

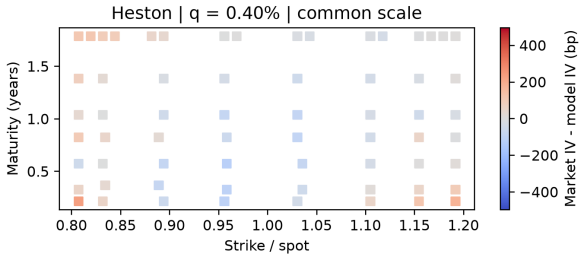
The residual maps, the parameters and the valuation paths come from one recalibration of the 2 September sample with $q = 0.40\%$, dated 24 October 2026. The Treasury curve and the gold spot anchor are dated 2 September as well.

5.2 What the residual maps add

The residual heatmaps show where each specification systematically over- or under-prices the observed implied-volatility surface. Black–Scholes leaves the strongest smile and term-structure pattern. Bates is refined from the nested Heston point and Bates–Hawkes from the nested Bates point, so each richer model fits at least as well as the model it contains (Table 2). All maps use one colour scale; calibration fit does not establish economic superiority.

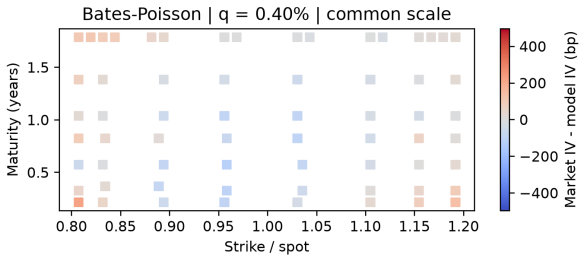


(a) Black–Scholes

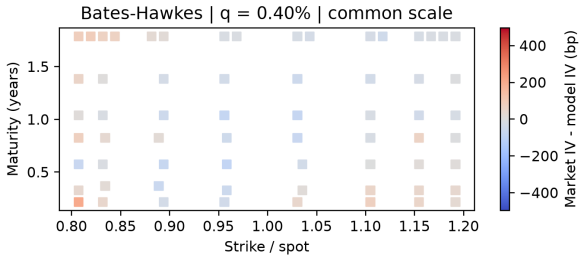


(b) Heston

Figure 6. Implied-volatility residual maps for the common 2 September CC64 sample with $q = 0.40\%$. Blue denotes model IV above market IV and red denotes model IV below market.



(c) Bates–Poisson



(d) Full Bates–Hawkes

Figure 6 (continued). Bates–Poisson and Full Bates–Hawkes residual maps; all colourbars share the same residual range.

Table 2. Current-surface IV calibration RMSE (bp). The 7 September column used $q = 0$; the IV targets differ, so the columns are not a like-for-like fit comparison.

Model	$q = 0.40\%$	7 Sep, $q = 0$
Black–Scholes	131.58	97.21
Heston	68.15	49.49
Bates–Poisson	65.24	49.41
Full Bates–Hawkes	52.83	49.33

5.3 What the fit can and cannot identify

With the fee as carry the fit deteriorates: the Heston price RMSE rises from USD 0.49 to 0.75, and the loss increases in every moneyness bucket. The calls therefore price a higher carry than Treasury rates less the fee. Call-only surfaces cannot separate that carry from skew, so a fitted carry would not be a forward; the futures-implied fee is used instead, and identifying the forward would need puts or box-spread rates.

Calibration fragility

The Feller condition is imposed as a hard constraint and binds at the optimum for Bates and Bates–Hawkes; without it the Heston loss falls by 68%. The Heston loss changes by less than 3% for κ between 2 and 4, so mean reversion is weakly identified. With $q = 0.40\%$ the Bates jump intensity is 1.72 a year rather than zero, but jump and Hawkes parameters rest on 64 calls and are not individually identified. Bates–Hawkes branching is 0.825 with a kernel half-life of 24.7 days.

6 DCF DRIVER AND TERMINAL-VALUE AUDIT

Growth, reinvestment, WACC and terminal closure are shared across gold engines. Table 3 summarizes the corporate assumptions.

6.1 Growth, ROIC and reinvestment

Growth and ROIC converge linearly to their stable levels over the five rate years since valuation. Reinvestment follows $b_t = g_t/ROIC_t$, linking growth to the capital required to sustain it (Figure 7).

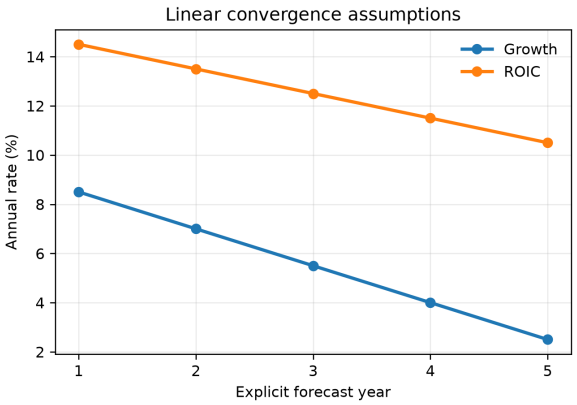


Figure 7. Common explicit-period DCF schedule: growth and ROIC converge linearly over the five-year horizon.

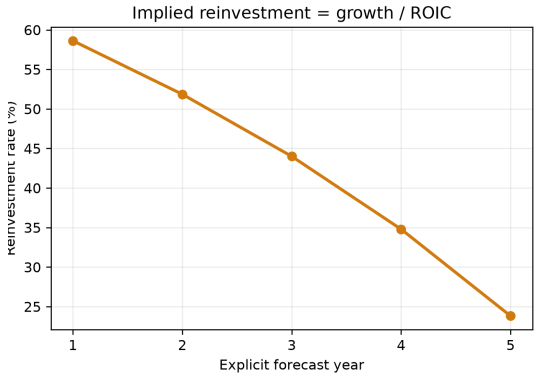


Figure 7 (continued). The implied reinvestment rate, defined as growth divided by ROIC, falls from 58.62% to 23.81%.

6.2 Common stochastic WACC

WACC follows the bounded mean-reverting process in Equation (18), with inputs in Table 3; w_0 applies to the first year after valuation. Its annual innovation scale is $\sigma_w^* = \sigma_w\sqrt{(1 - e^{-2\kappa_w})/(2\kappa_w)} = 0.7308$ percentage points before clipping. All engines reuse the same 8,192 shock paths (Figure 8).

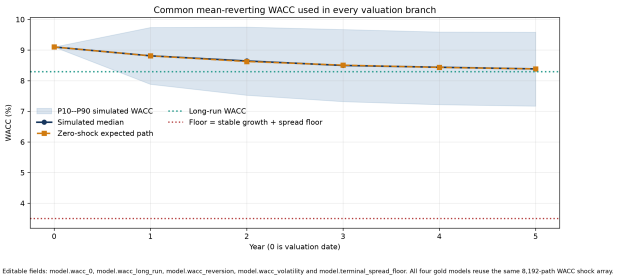


Figure 8. Common mean-reverting WACC. The band is the P10–P90 range from the shared 8,192-path shock array; the orange line is the zero-shock path and the dotted lines show the long-run level and admissibility floor.

6.3 Terminal closure and sensitivity

The terminal block uses $g_\infty = 2.50\%$, $ROIC_\infty = 10.50\%$ and terminal reinvestment of 23.81%. At the median terminal WACC of 8.445%, the terminal multiple is 13.14× the final after-tax margin. The spread floor enforces $w_H > g_\infty$; 0.00% of simulated rates reach a clipping bound.

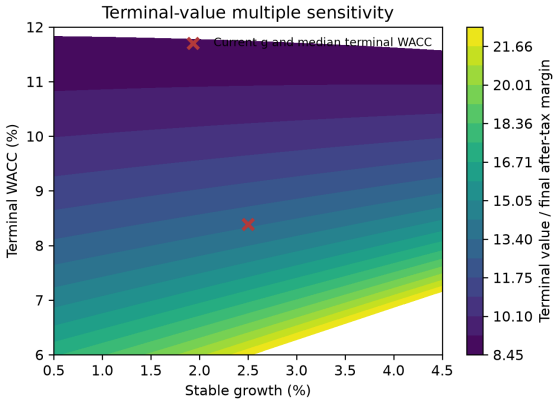


Figure 9. Terminal-value multiple sensitivity to stable growth and terminal WACC. The cross marks the configured stable-growth rate and current median terminal WACC.

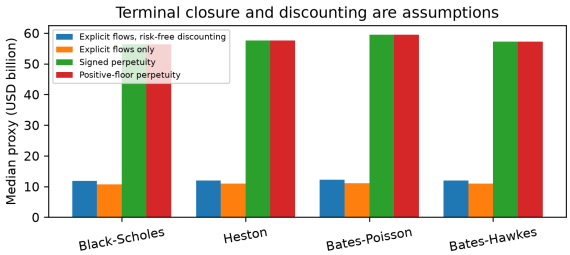


Figure 9 (continued). Median proxy under the terminal conventions and with risk-free discounting of the explicit flows. Explicit flows to 2030 are worth about USD 11 billion against about USD 57–59 billion with a perpetuity. Zero terminal is a truncation scenario, not an estimated reserve life.

Table 3. Editable common DCF inputs.

Parameter	Value
Tax rate	25.00%
$g_1 \to g_\infty$	8.50–2.50%
$ROIC_1 \to ROIC_\infty$	14.50–10.50%
$WACC_0 \to \bar{w}$	9.10–8.30%
WACC reversion	0.45
WACC volatility	0.90%
WACC floor/cap	3.50/25.00%
Terminal reinvestment	23.81%

Interpretation

These are methodological assumptions, not filing-calibrated Barrick estimates. Terminal closure dominates the reported operating proxy: it accounts for about 83% of the mean. The former positive floor on the final margin is no longer used; it leaves medians unchanged.

The quarterly bridge multiplies the average simulated gold price less unit Cost of Sales, both in USD per ounce, by production in thousand ounces, then divides by 1,000 to obtain component margin in USD millions. Each quarter is taxed, charged its reinvestment and discounted at its own date. This proxy is not reported EBIT or FCFF: production must be reconciled with sales and inventory, and depreciation, capital expenditure and working capital must enter without double counting. Fixed operating vectors exclude mine-level shocks and price-cost dependence.

7 OPERATING-PROXY SENSITIVITIES

Table 4. Bates–Hawkes operating proxy, USD billion unless stated.

Metric	Value
P10	1.26
Median (P50)	57.26
P90	155.88
Mean	73.11
Negative proxy paths	9.3%
Explicit flows to 2030, median	10.90
Same, risk-free discounting	11.97

Interpretation

These are unconstrained aggregate operating proxies, not enterprise or equity values. Negative outcomes are not negative limited-liability share prices, and no market comparison is made.

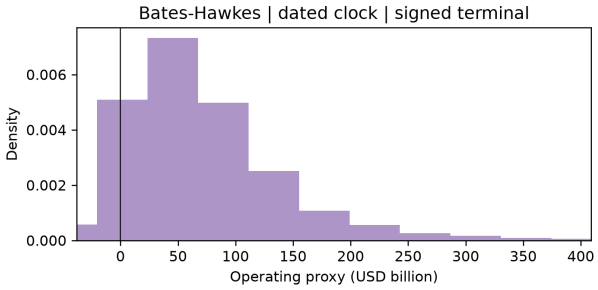


Figure 10. Signed-terminal operating proxy under Bates–Hawkes on the dated clock, 8,192 paths. Display limits are the 0.5th–99.5th percentiles; the full tails remain in the data. This is not a confidence interval or a physical probability law.

7.1 Model choice matters less than numerical noise

Medians range from USD 56.32 to 59.48 billion across the four engines and P90 from 155.88 to 160.19. Doubling the paths to 16,384 or changing the seed moves single-model medians across USD 56.24–59.99 billion; for Bates–Hawkes alone the range is 56.26–58.42. The numerical range of one model is as wide as the spread between models, so the four medians do not rank the models. Explicit flows to 2030 are worth USD 10.75–11.08 billion and the terminal value carries about 83% of the mean.

Table 5. Proxy medians, USD billion. Explicit: flows to 31 December 2030 without terminal value.

Gold engine	Median	Explicit
Black–Scholes	56.32	10.75
Heston	57.62	10.90
Bates–Poisson	59.48	11.08
Full Bates–Hawkes	57.26	10.90

7.2 What the corrections changed

Table 6 applies the corrections of this revision one at a time to the 7 September medians. The dated clock removes Barrick's first-half 2026 margin, which the earlier version discounted as a future flow, and ends the explicit horizon at 2030; between the two editions the explicit flows fall from about USD 13 to 11 billion. Each step moves the medians by less than the numerical range above.

Table 6. Median proxy, USD billion, as each correction is added.

Step	BS	Heston	Bates	B–H
7 Sep Paper	60.10	59.77	59.12	59.82
+ dated clock	57.81	58.65	59.34	58.49
+ spot anchor	57.11	57.95	58.59	57.74
+ GLD carry	56.32	57.62	59.48	57.26

7.3 Structural illustration, not model selection

Bates–Hawkes illustrates the richest law, not a preferred corporate valuation. Its branching ratio is 0.825 and its kernel half-life $\log(2)/\beta$ is 24.7 days. These are risk-neutral fitted parameters, not measured geopolitical persistence.

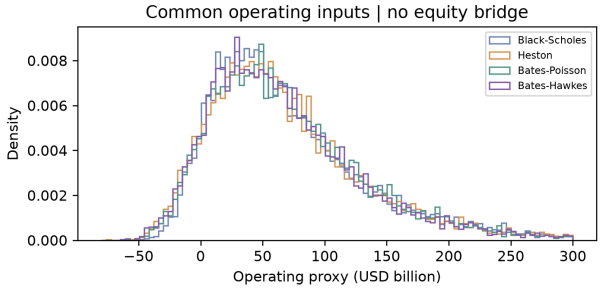


Figure 11. Operating-proxy distributions under four gold-price laws. Operations, WACC and signed terminal closure are common. No share count, equity bridge or market reference is used.

8 COPPER EXTENSION: HISTORY AND PAIRED OPERATING VALUES

8.1 Perimeter and units

The original gold-only study omitted copper. The extension admits three producing mines: Lumwana (100%), Zaldívar (50%) and Jabal Sayid (50%). Their Q2 2026 attributable sales are 40, 9 and 5 kt, and Cost of Sales is 3.14, 4.89 and 2.63 USD/lb, respectively (Barrick, 2026b). The sales quantities already incorporate ownership; multiplying by those shares again would understate the two joint ventures. Lumwana is consolidated, while the other two interests are equity-accounted (Barrick, 2025). Production and sales are distinct quantities; the copper margin uses sales.

With s_i in kt and P_t, c_i in USD/lb, quarterly attributable component margin in USD millions is

$$CM_t^{Cu} = \frac{2204.6226218488}{1000} \sum_i s_i (P_t - c_i). \tag{21}$$

Q2 sales and costs are held fixed for the quarters valued, from the 2026Q3 stub to 2030Q4. This is a controlled operating scenario, not a mine-production forecast. CoS includes depreciation; reported capital expenditure is not separately deducted in this proxy. Expansion at Lumwana, Reko Diq, other gold assets, reserves and closure obligations remain outside the model.

8.2 Historical surface and exercise treatment

The IB API collection selected ten historical dates between 9 July and 2 September 2026 before scoring. Each date used 112 uniformly spaced out-of-the-money contract locations, rather than thousands of quotes. There are 912 returned quote rows across the pilot; missing responses are not fabricated. The common admission rule requires at least 64 eligible observations and three expiries, option and matched-future quote ages no greater than 120 seconds, and a relative option spread no greater than 20%. Seven dates survive, creating six origin–target pairs with gaps of 11, 23, 9, 10, 1 and 1 calendar days. July 10, July 31 and August 3 fail the quality gate. Historical data therefore exist, but this is a sparse pilot rather than a daily panel; the other gold reference dates were not collected in this experiment.

Copper is quoted in USD/lb through COMEX HG futures and their American options. A 600-step constant-volatility CRR inversion removes the early-exercise convention before fitting European forward-price engines; a 300-step comparison checks discretization. This transformation does not solve early exercise under the full stochastic-volatility/jump models. Each option is matched to its own underlying

future; Black–76 is the constant-volatility benchmark. Origin calibration uses the dated Treasury curve, with no target-date quote entering parameter selection. The 2 September current fit uses all 86 eligible observations.

8.3 Temporal tests

Origin parameters are frozen and their latent states projected over the actual calendar gap. Target forward prices and rates condition repricing; the test is not an unconditional physical-price forecast. All four models share 480 target observations. Persistence uses origin IV interpolation inside the maturity–log-moneyness convex hull and has 375 supported targets. Equal-date scoring prevents dense dates from dominating; unsupported persistence points are omitted from its comparison.

Table 7. Copper rolling temporal repricing, six date pairs. RMSE is the mean daily RMSE in IV basis points; R^2_p is against observed-IV persistence on common support.

Model	RMSE (bp)	R^2_p
Black–76	106.48	−1.294
Heston	71.46	−0.422
Bates	70.49	−0.389
Bates–Hawkes	67.65	−0.208

No model beats persistence in the rolling experiment. As a robustness exercise, fixing 9 July as the sole calibration origin gives persistence R^2 of −0.345, −0.282, 0.193 and 0.182 in the same model order, on 333 supported targets. The positive Bates and Hawkes point estimates are conditional on this different origin, gap and support; they do not overturn the rolling result or establish significance with six dependent target dates.

8.4 Paired contribution to 2030

Under the same dated clock, the copper scenario uses log-linear matched-future anchors at actual delivery dates, flat beyond the final observed delivery at 0.728 years. This schedule is neither a fixed-futures drift nor a fitted physical expectation, so the horizon to 2030 requires substantial extrapolation. Copper paths use the same corrected daily engines as gold. The gold parameters, operating vectors and WACC paths are reused exactly for gold-only and combined runs. The primary case assumes independent commodity paths and no terminal value. Whole-path permutations with terminal Gaussian-score correlation ±0.5 and a signed perpetuity are separate sensitivities, not estimated price dependence or reserve-supported continuation. Copper Bates fits zero jump intensity, so that engine reduces to Heston.

Table 8. Paired means of explicit flows to 2030, USD bn. Δ is the mean pathwise copper increment; SE is its Monte Carlo standard error in USD mn. These are operating proxies.

Model	Gold	Gold+Cu	Δ	SE
Black benchmark	12.060	14.368	2.308	17.9
Heston	11.981	14.285	2.304	28.5
Bates	12.062	14.379	2.317	31.5
Bates–Hawkes	12.154	14.484	2.330	28.1

The additional margin enters the same tax and growth–ROIC operator before discounting. The copper increment is about USD 2.30–2.33 bn under this contract; it cannot be added to the gold signed-terminal medians reported earlier, which use a different statistic and horizon. Monte Carlo standard errors measure simulation noise only. Mine-level forecast, commodity basis, accounting and parameter uncertainty remain outside them. The extension closes an omitted operating component while leaving the corporate FCFF and equity bridges open.

9 OUT-OF-SAMPLE EVIDENCE AND MODEL RISK

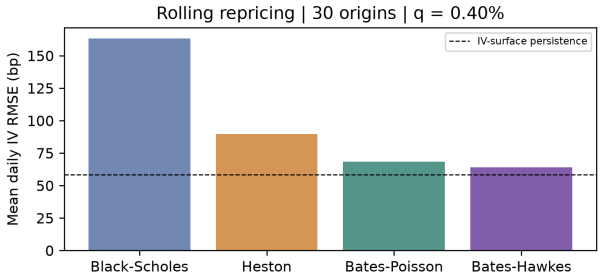


Figure 12. Rolling repricing on 30 origins and 4,667 common targets, $q = 0.40\%$, same search policy for every model. The dashed line is the error of IV-surface persistence on its support.

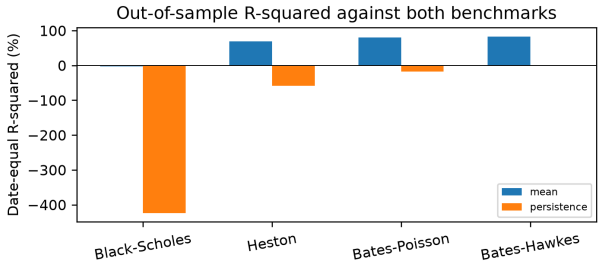


Figure 12 (continued). Equal-date R^2_{OOS} against the origin mean IV and against persistence; positive values indicate improvement over the benchmark.

9.1 Conditional next-date repricing

Each origin is recalibrated on its own CC64 sample with a global search for every model; no model starts from an earlier origin. Model comparisons use 4,667 targets; persistence comparisons use the 4,026 inside the origin interpolation domain, so the two benchmarks have different supports. Equal date weighting limits the influence of densely populated dates.

The mean daily IV RMSE is 163.35 bp for Black–Scholes, 89.62 for Heston, 68.39 for Bates and 64.16 for Bates–Hawkes. Against the origin mean IV the equal-date R^2_{OOS} is -2.33%, 69.17%, 80.62% and 83.03%. Against persistence it is -423.92%, -59.00%, -17.47% and -1.84%: no model beats carrying the previous surface forward, and Bates–Hawkes is better than persistence on 13 of 30 dates.

Model-risk conclusion

Diebold–Mariano tests on daily MSE with Newey–West variance (Diebold and Mariano, 1995; Newey and West, 1987) rank Bates–Hawkes above Black–Scholes and Heston ($p < 0.001$), but not above Bates ($p = 0.165$) and not away from persistence ($p = 0.772$). With 30 dependent dates the richest model is the best of the four, not a demonstrated improvement on the simplest benchmark.

9.2 Jump-layer interpretation

The self-exciting extension is motivated by clustered option-implied jump activity: relative to a constant Poisson rate, the Hawkes intensity permits bursts followed by endogenous decay. The fitted branching ratio of 0.825 and half-life of 24.7 days are risk-neutral parameters, not evidence of physical news clustering; without a dated news filtration or parameter-stability evidence, these bursts cannot be identified as responses to geopolitical events.

9.3 Interpretive limitations and validation

Reproducibility, numerical accuracy and economic validity are distinct. A good option fit cannot validate the GLD-to-gold mapping or the scenario/WACC combination. The experiment identifies sensitivity to a chosen numerical contract, not corporate fair value.

No per-share value is reported: debt, cash and other claims are not reconciled. A genuine valuation must start from one dated filing and the matching listed security, establish ownership and attributable production, and distinguish component margin from reported operating profit. Depreciation, cash taxes, working capital, sustaining capital expenditure and growth investment must then enter consistently, with no deduction counted twice. Net debt, non-operating assets, streams

and other claims must be matched to a diluted share count on the same basis.

9.4 Operating uncertainty and research priorities

Production, grade, recovery, mine mix and costs may depend on gold prices, inflation and exchange rates; operating dependence requires explicit assumptions. A mining-company terminal block needs an explicit resource interpretation: continued production may require reserve conversion, replacement investment or new projects, while depletion and closure liabilities shorten the cash-flow stream. A positive WACC-growth spread controls a denominator but does not justify a perpetual operating horizon. Market maturities end at 653 days, well before 2030 and the perpetuity, so fixed parameters beyond that horizon are an extrapolation assumption. A finite-reserve model and a bucketed term structure remain necessary for an economic valuation.

CONCLUSION

Holding operations and corporate assumptions fixed isolates the effect of gold-price model choice. On a valuation clock that counts only flows after 2 September 2026, the four option-implied laws give median proxies of USD 56.32–59.48 billion, inside the Monte Carlo range of a single model, while the perpetuity after 2030 carries about 83% of the mean. The model choice therefore matters less for this proxy than numerical precision and, above all, the terminal assumptions.

In rolling repricing Bates–Hawkes has the lowest error but is statistically indistinguishable from Bates and from IV-surface persistence. Treating the GLD fee as carry is economically required and corroborated by gold futures, yet it worsens the in-sample fit, which signals that call-only surfaces cannot identify the forward. The copper extension adds about USD 2.30–2.33 billion of explicit flows; its sparse rolling experiment does not beat persistence either.

The contribution is an auditable interdisciplinary framework, not a certified value. The next required results are a validated physical commodity forecast linked to Barrick's realised prices, a forward identified from puts or box spreads, stochastic operating inputs, a reconciled FCFF and equity bridge, and a finite-resource terminal closure.

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